CHAPTER 9

Matrices

Operations, inverse, powers, systems and transformations of the plane

Objectives. Compute with matrices (sum, product by a real number, product); find the inverse of a matrix of order 2, and of order 3 with a calculator; compute a power, in particular by induction; solve a linear system in matrix form; represent a transformation of the plane by a matrix.

Prerequisites. Linear systems of two equations; coordinates of vectors; addition formulas (chapter 3); proof by induction; sequences.

Fluency

Fluency · No calculator, five minutes.

1. Solve the system $3x + 5y = 4$ and $x + 2y = 1$.
2. Compute the determinant of the vectors $\overrightarrow{a}\begin{pmatrix}3\\1\end{pmatrix}$ and $\overrightarrow{b}\begin{pmatrix}5\\2\end{pmatrix}$.
3. Give $\cos\frac{\pi}{3}$ and $\sin\frac{\pi}{3}$.
4. What are the coordinates of the image of the point (x, y) under the reflection in the axis (Ox)?
5. Expand $(a+b)^2$. Do we use $ab = ba$?
6. Let $u_{n+1} = 2u_n$ and $u_0 = 3$. Express u_n in terms of n.

Solution · Fluency.

1. $x = 3$ and $y = -1$.
2. $3 \times 2 - 1 \times 5 = 1$.
3. $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$.
4. $(x, -y)$.
5. $a^2 + 2ab + b^2$: we group $ab + ba$ into $2ab$, which uses $ab = ba$. This will no longer be possible with matrices.
6. $u_n = 3 \times 2^n$.

Introductory activity

Activity 1 · Turning and stretching a figure 40 min . We represent a point $M(x,y)$ by the column $X = \begin{pmatrix} x \\ y \end{pmatrix}$. For a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we define the product $AX = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}$: it is the column of a new point.

1. Let $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Compute the images under R of the vertices of the triangle $O(0,0)$, $P(2,0)$, $Q(2,1)$. Draw both triangles. Which transformation do you recognise?
2. Let $E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Same question.
3. Compute $R(EX)$ for any $X = \begin{pmatrix} x \\ y \end{pmatrix}$. Find a matrix C such that $R(EX) = CX$ for every X.
4. Compute likewise $E(RX)$ and the corresponding matrix. Are the two composite transformations the same? Check on the point Q.
5. Suggest a rule for computing C directly from R and E: it is the matrix product RE.

Observing is not proving: the properties conjectured here are established in the course.

Solution · Activity.

1. $O \mapsto O$, $P \mapsto (0,2)$, $Q \mapsto (-1,2)$: rotation with centre O and angle $\frac{\pi}{2}$.
2. $O \mapsto O$, $P \mapsto (4,0)$, $Q \mapsto (4,1)$: stretch with factor 2 parallel to the horizontal axis.
3. $EX = \begin{pmatrix} 2x \\ y \end{pmatrix}$ then $R(EX) = \begin{pmatrix} -y \\ 2x \end{pmatrix}$, so $C = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$.
4. $RX = \begin{pmatrix} -y \\ x \end{pmatrix}$ then $E(RX) = \begin{pmatrix} -2y \\ x \end{pmatrix}$, with matrix $\begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$. The transformations differ: Q has image $(-1,4)$ in the first case and $(-2,2)$ in the second.
5. The entry in row i and column j of C is obtained by multiplying row i of R by column j of E, term by term, and adding. The order matters: $RE \neq ER$.

Teacher note · Running the activity. Have the figures drawn on squared paper or with software. Non-commutativity should appear as a geometric observation (stretch then turn, or turn then stretch) before it becomes a rule of computation.

Course notes

Vocabulary and operations

Definition · Matrix. A *matrix* of size $n \times p$ is an array of real numbers with n rows and p columns. The entry in row i and column j of A is written a_{ij}. An $n \times 1$ matrix is a *column* matrix, a $1 \times p$ matrix a *row* matrix, an $n \times n$ matrix a *square* matrix of order n. The *identity* matrix I_n has 1s on the diagonal and 0s elsewhere; the *zero* matrix has only 0s. A square matrix is *diagonal* if its entries off the diagonal are zero.

Definition · Sum, product by a real number, product. For A and B of the same size and k real, $A + B$ and kA are computed entry by entry.

If A has size $n \times p$ and B has size $p \times q$, the product AB is the $n \times q$ matrix with entries

$$c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj}.$$

Example 1 · A product. $\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} 1\times 0 + 2 \times 4 & 1 \times 1 + 2 \times (-2) \\ -1 \times 0 + 3 \times 4 & -1 \times 1 + 3 \times (-2) \end{pmatrix} = \begin{pmatrix} 8 & -3 \\ 12 & -7 \end{pmatrix}$.

Property · Rules of computation. When the products are defined: $(AB)C = A(BC)$, $A(B + C) = AB + AC$, $(A + B)C = AC + BC$, $(kA)B = k(AB)$, and $AI_n = I_n A = A$ for A square of order n.

Watch out.

1. In general $AB \neq BA$; consequently $(A + B)^2 = A^2 + AB + BA + B^2$, and not $A^2 + 2AB + B^2$.
2. A product can be zero without either factor being zero: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

Powers

Definition · Powers of a square matrix. For A square of order n, let $A^0 = I_n$ and $A^{k+1} = A^k A$ for every natural number k.

Method · Conjecture, then prove by induction. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, we compute $A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $A^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ and conjecture $A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$. This is true for $k = 0$; if it is true for k, then $A^{k+1} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$.

Key point · Diagonal matrix. If D is diagonal with diagonal entries $d_1, \ldots, d_n$, then D^k is diagonal with entries $d_1^k, \ldots, d_n^k$.

Inverse and systems

Definition · Invertible matrix. A square matrix A of order n is *invertible* if there is a matrix B such that $AB = BA = I_n$. This matrix is unique; it is written A^{-1}.

Property · Inverse of a matrix of order 2. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The matrix A is invertible if and only if $ad - bc \neq 0$, and then

$$A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The number $ad - bc$ is the *determinant* of A, written $\det A$.

Proof. Let $M = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Computation gives $AM = MA = (ad - bc)I_2$. If $ad - bc \neq 0$, the matrix $\frac{1}{ad-bc}M$ is therefore the inverse of A.

If $ad - bc = 0$, then $AM = 0$. Suppose A is invertible: multiplying on the left by A^{-1} gives $M = 0$, so $a = b = c = d = 0$ and A is zero. But the zero matrix is not invertible ($0 \times B = 0 \neq I_2$): a contradiction. □

Property · Linear systems. A linear system of n equations in n unknowns can be written $AX = B$, where A is the matrix of coefficients, X the column of unknowns and B the column of right-hand sides. If A is invertible, the system has a unique solution: $X = A^{-1}B$.

Proof. If $AX = B$, then $A^{-1}AX = A^{-1}B$, that is $X = A^{-1}B$. Conversely, $A\left(A^{-1}B\right) = B$. □

Method · Order 3: calculator and check. For a matrix of order 3, obtain A^{-1} with a calculator or software, then check one or two entries of AA^{-1} by hand.

Transformations of the plane and sequences

Property · Matrix of a rotation. The rotation with centre O and angle θ maps the point with column X to the point with column $R_\theta X$, where

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Proof. The point $M(x, y)$ has affix $z = x + iy$, and its image has affix $e^{i\theta}z = (\cos\theta + i\sin\theta)(x + iy) = (x\cos\theta - y\sin\theta) + i(x\sin\theta + y\cos\theta)$ (chapter 3): these are the entries of $R_\theta X$. □

Method · Sequence defined by a matrix. If $X_{k+1} = AX_k$ for every k, then $X_k = A^k X_0$ (immediate induction). A sequence defined by $u_{k+2} = au_{k+1} + bu_k$ reduces to this with $X_k = \begin{pmatrix} u_{k+1} \\ u_k \end{pmatrix}$ and $A = \begin{pmatrix} a & b \\ 1 & 0 \end{pmatrix}$.

Exercises

Exercise 9.1 · Computations. Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 4 & -2 \end{pmatrix}$. Compute $A+B$, $2A-B$, AB and BA.

Solution. $A+B = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$, $2A-B = \begin{pmatrix} 2 & 3 \\ -6 & 8 \end{pmatrix}$, $AB = \begin{pmatrix} 8 & -3 \\ 12 & -7 \end{pmatrix}$, $BA = \begin{pmatrix} -1 & 3 \\ 6 & 2 \end{pmatrix}$. We see that $AB \neq BA$.

Exercise 9.2 · Inverse and system of order 2. Let $A = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$.

1. Explain why A is invertible and find A^{-1}.
2. Deduce the solution of the system $3x+5y=4$, $x+2y=1$.

Solution.

1. $\det A = 6-5 = 1 \neq 0$ and $A^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$.
2. $X = A^{-1}\begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 8-5 \\ -4+3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$: $x=3$, $y=-1$.

Exercise 9.3 · System of order 3. Consider the system $x+y+z=6$, $2x-y+z=3$, $x+2y-z=2$.

1. Write it in the form $AX=B$.
2. With a calculator, find A^{-1}, then the solution of the system.
3. Check the solution by hand.

Solution.

1. The system can be written $AX=B$ with
$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{pmatrix}, \qquad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad B = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix}.$$
2. The calculator gives
$$A^{-1} = \frac{1}{7}\begin{pmatrix} -1 & 3 & 2 \\ 3 & -2 & 1 \\ 5 & -1 & -3 \end{pmatrix} \qquad \text{and} \qquad X = A^{-1}B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$
3. $1+2+3=6$, $2-2+3=3$, $1+4-3=2$.

Exercise 9.4 · Powers by induction. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Compute A^2 and A^3, conjecture an expression of A^n for $n \geq 1$, then prove it by induction.

Solution. $A^2 = \begin{pmatrix} 4 & 4 \\ 0 & 4 \end{pmatrix}$, $A^3 = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix}$. Conjecture: $A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$. True for $n = 1$. If it holds for n:

$$A^{n+1} = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2^{n+1} & 2^n + n2^n \\ 0 & 2^{n+1} \end{pmatrix},$$

and $2^n + n2^n = (n+1)2^n$: the formula holds for $n+1$.

Exercise 9.5 · A polynomial relation. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

1. Check that $A^2 = 2A + 3I_2$.
2. Deduce that A is invertible and express A^{-1} in terms of A and I_2. Check with the formula of the course.

Solution.

1. $A^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ and $2A + 3I_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.
2. $A^2 - 2A = 3I_2$ can be written $A \times \frac{1}{3}(A - 2I_2) = I_2$, and likewise $\frac{1}{3}(A - 2I_2)A = I_2$. So $A^{-1} = \frac{1}{3}(A - 2I_2) = \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$. With the course: $\det A = -3$ and $A^{-1} = -\frac{1}{3}\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$, the same result.

Exercise 9.6 · Rotations.

1. Using the addition formulas, prove that $R_\theta R_{\theta'} = R_{\theta+\theta'}$.
2. Deduce $(R_{\pi/6})^{12}$.
3. Find the image of the point $(2, 0)$ under the rotation with centre O and angle $\frac{\pi}{3}$.

Solution.

1. The entry in row 1 and column 1 of the product is $\cos\theta\cos\theta' - \sin\theta\sin\theta' = \cos(\theta + \theta')$; the entry in row 2 and column 1 is $\sin\theta\cos\theta' + \cos\theta\sin\theta' = \sin(\theta + \theta')$. The other two are obtained in the same way.
2. By induction, $(R_{\pi/6})^{12} = R_{2\pi} = I_2$.
3. $R_{\pi/3}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\cos\frac{\pi}{3} \\ 2\sin\frac{\pi}{3} \end{pmatrix} = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$: the point $(1, \sqrt{3})$.

Exercise 9.7 · A parabola through three points. Find the real numbers a, b, c such that the parabola with equation $y = ax^2 + bx + c$ passes through the points $(1,2)$, $(2,3)$ and $(3,6)$. Write the system in matrix form.

Solution. The system is

$$\begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}.$$

Subtracting consecutive equations: $3a + b = 1$ and $5a + b = 3$, so $a = 1$, $b = -2$, then $c = 3$. Check: $1 - 2 + 3 = 2$, $4 - 4 + 3 = 3$, $9 - 6 + 3 = 6$.

Problem

Problem · The Fibonacci sequence and Cassini's identity. The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for every $n \geq 0$. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Compute $F_2, \ldots, F_6$ and A^2, A^3.
2. Prove by induction that, for every $n \geq 1$, $A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$.
3. Prove that for all matrices M and N of order 2, $\det(MN) = \det M \times \det N$. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$.
4. Deduce that $\det(A^n) = (-1)^n$, then Cassini's identity: $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ for $n \geq 1$.
5. Check the identity for $n = 5$.
6. Using $A^{2n} = A^n \times A^n$, prove that $F_{2n} = F_n(F_{n+1} + F_{n-1})$ for $n \geq 1$.

Beyond the syllabus · Determinant of a product. The relation $\det(MN) = \det M \times \det N$ of question 3 is not part of the syllabus. It is proved here by computation for matrices of order 2 and is not assessed.

Solution · Problem.

1. $F_2 = 1$, $F_3 = 2$, $F_4 = 3$, $F_5 = 5$, $F_6 = 8$. $A^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $A^3 = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$.
2. For $n = 1$: $\begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = A$. If the formula holds for n, then
$$A^{n+1} = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_{n+1}+F_n & F_{n+1} \\ F_n + F_{n-1} & F_n \end{pmatrix} = \begin{pmatrix} F_{n+2} & F_{n+1} \\ F_{n+1} & F_n \end{pmatrix}.$$
3. $MN = \begin{pmatrix} aa'+bc' & ab'+bd' \\ ca'+dc' & cb'+dd' \end{pmatrix}$. Its determinant is $(aa'+bc')(cb'+dd') - (ab'+bd')(ca'+dc')$. After expanding, the terms $aa'cb'$ and $bc'dd'$ cancel with their opposites, and what remains is $ada'd' + bcb'c' - adb'c' - bca'd' = (ad-bc)(a'd'-b'c')$.
4. By induction, $\det(A^n) = (\det A)^n = (-1)^n$. With question 2, $\det(A^n) = F_{n+1}F_{n-1} - F_n^2$.
5. $F_6F_4 - F_5^2 = 24 - 25 = -1 = (-1)^5$.
6. The entry in row 1 and column 2 of A^{2n} is F_{2n}; that of A^nA^n is $F_{n+1}F_n + F_nF_{n-1} = F_n(F_{n+1}+F_{n-1})$.

Teacher note · Status of the problem. The determinant of order 2 is used for the inverse; its multiplicativity is not in the syllabus and is proved here by computation. Possible extension: the Hill cipher, which encrypts pairs of letters with a matrix of order 2 invertible modulo 26, listed among the possible problems of the syllabus.